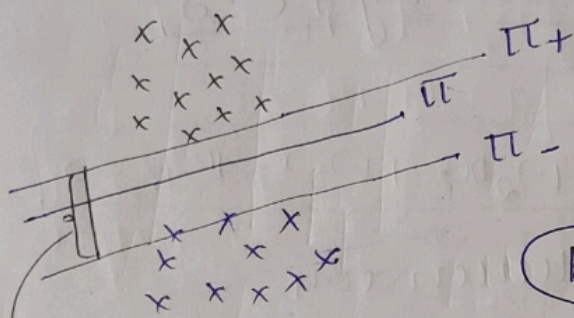


2. Support Vector machine

1. Geometric Intuition



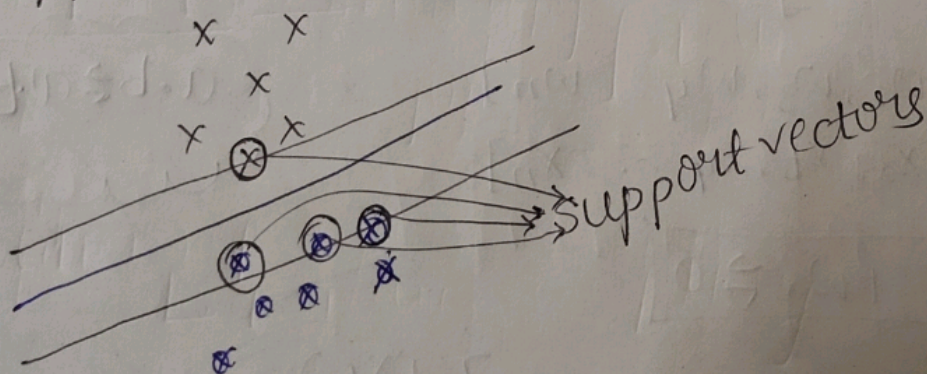
Kernels

- ⊕ Robust to outliers
- ⊕ also work on non-linear data
- ⊕ used in both Regression & classification

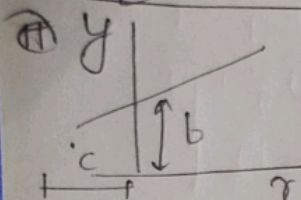
distance called margin = d

to Algo: select the such π which has d is maxim

① support vector



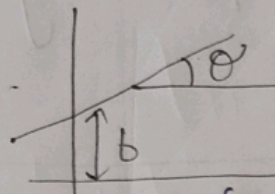
Equation of Hyperplane

①  $y \hat{=} mx + c$

general form =
 $ax + by + c = 0$
 $y = -\frac{a}{b}x - \frac{c}{b}$

$m = -\frac{a}{b}$ $c = -\frac{c}{b}$

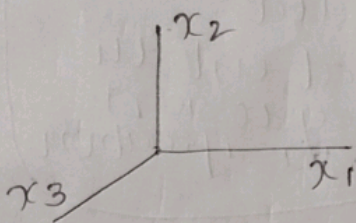
generalizing
 #2 lines


 $ax + by + c = 0$

$w_1 x_1 + w_2 x_2 + w_0 = 0$

or
 $ax + by + c = 0$

② 3 lines

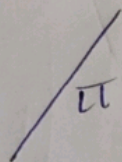


$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$

for n dimension
 so,

$w_1 x_1 + w_2 x_2 + \dots + w_n x_n + w_0 = 0$

③ Hyperplane represented by π



and represent points in vectors

now represent in the vector
 for clear calculation

$w = [w_1, w_2, w_3, w_4 \dots w_n]$

$x = [x_1, x_2, x_3, x_4 \dots x_n]$

$a \cdot b \hat{=} a^T b$

④ $w \cdot x + w_0 \hat{=} 0$

$[w_1, w_2, w_2, w_3 \dots w_n] \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_n \end{bmatrix}$

$$\rightarrow w \cdot x + w_0 = 0$$

$$\boxed{w^T x + w_0 \geq 0}$$

$$\therefore a \cdot b = a^T b$$

assume Hyperplane passed from origin
then $w_0 = 0$

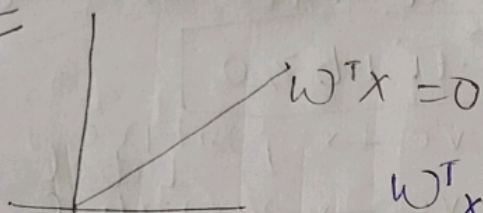
So,

$$\boxed{w^T x = 0}$$

for 2D $= w_1 x_1 + w_2 x_2 = 0$

4D $= w_1 x_1 + w_2 x_2 + \dots + w_3 x_3 + w_4 x_4 = 0$

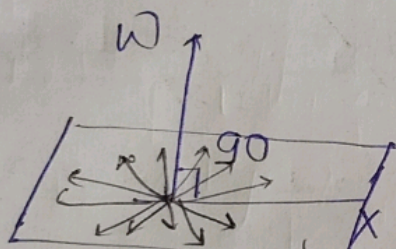
#



$$w^T x = w \cdot x = \|w\| \|x\| \cos \theta = 0$$

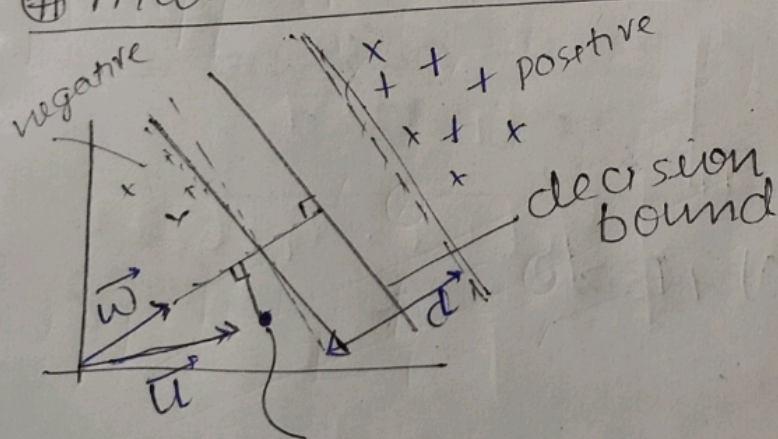
$$\theta = 90^\circ \text{ then } \cos 90^\circ = 0$$

w is $\perp$ to hyper-plane



hyperplane.

⊕ mathematical derivation



⊕ check whether

⊕ make a decision
Rule

to check the point
lies on +ve or -ve

$$\text{So, } \vec{w} \cdot \vec{u} > c$$

$$\Rightarrow \vec{w} \cdot \vec{u} - c > 0$$

$$\boxed{\vec{w} \cdot \vec{u} + b > 0}$$

+ve / -ve

eg: (i)

$$2x + 3y + 3 > 0$$

$$w_1 x + w_2 y + b > 0$$

$$\text{So, } \vec{w} (2, 3)$$

b

$$w_1 = 2, w_2 = 3, b = 3$$

$$\vec{w} = (2, 3)$$

new point to check
that is

$$\vec{u} = (3, 2)$$

$$\text{So, } \vec{w} \cdot \vec{u} + b > 0$$

$$2 \times 3 + 3 \times 2 + 3 > 0$$

$$6 + 6 + 3 > 0$$

$$\boxed{15 > 0}$$

+ve side lies

$$(ii) \vec{u} = (-2, 0)$$

$$\text{So, } \vec{w} \cdot \vec{u} + b > 0$$

$$(2\hat{i} + 3\hat{j}) \cdot (-2\hat{i} + 0\hat{j}) + 3 > 0$$

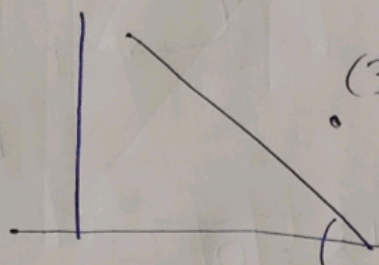
$$-4 + 0 + 3 > 0$$

$$\boxed{-1 > 0}$$

ve

-ve

so, it lies on
negative
side



(3, 2)

$$2x + 3y + 3 = 0$$

So find

general decision rule

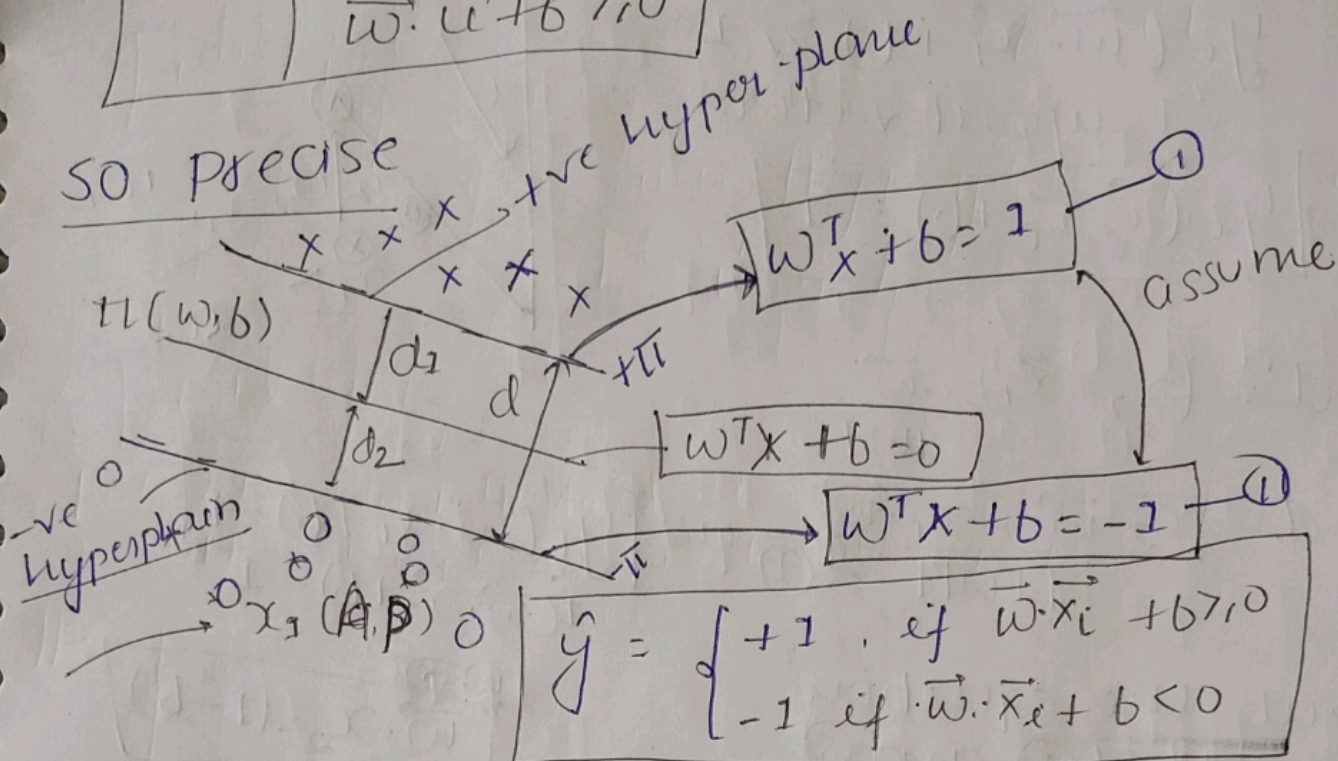
$$\boxed{\vec{w} \cdot \vec{u} + b > 0}$$

+ve $\rightarrow$ positive
label

-ve $\rightarrow$ -ve label

Decision Rule
 $\vec{w} \cdot \vec{x} + b \geq 0$

so precise



task
 → maximize the d using (w) and (b)

making equation of d

constraint 1

x hamesa $+1$ ke upar rahega so,

$$\vec{w} \cdot \vec{x} + b \geq 1 \quad \text{from eqn (i)}$$

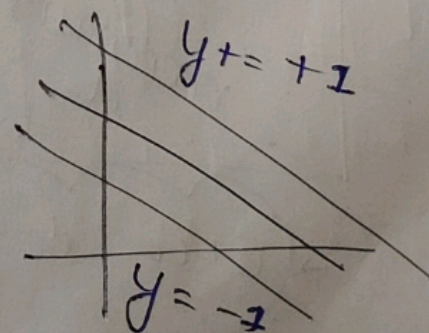
constraint - II

o hamesa -1 ke neeche rahega so,

$$\vec{w} \cdot \vec{x} + b \leq -1 \quad \text{eqn - (ii)}$$

combine both constrain.

y_i is the any points

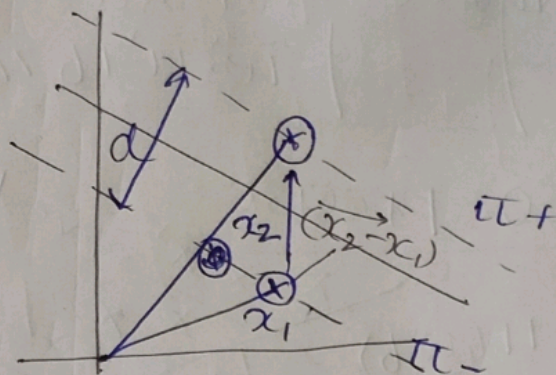


so combine both it

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$

or support vector:

$$y_i (\vec{w} \cdot \vec{x}_i + b) = 1$$



$$d = (\vec{x}_2 - \vec{x}_1) \cdot \frac{\vec{w}}{\|\vec{w}\|}$$

$$d = \frac{\vec{x}_2 \cdot \vec{w} - \vec{x}_1 \cdot \vec{w}}{\|\vec{w}\|} \quad \text{--- (7)}$$

$$\text{as } \begin{cases} y_i (\vec{w} \cdot \vec{x}_i + b) = 1 \\ 1(\vec{w} \cdot \vec{x}_2 + b) = 1 \end{cases}$$

$$\vec{w} \cdot \vec{x}_2 = (1 - b) \quad \text{--- (5)}$$

put in

$$\text{--- (4)} \quad -1(\vec{w} \cdot \vec{x}_1 + b) = 1$$

$$\vec{w} \cdot \vec{x}_1 = -b - 1 \quad \text{--- (6)}$$

$$d = \frac{[1 - b + b + 1]}{\|\vec{w}\|}$$

$$d = \frac{2}{\|\vec{w}\|}$$

$$\text{distance} = \frac{2}{\|\vec{w}\|}$$

$$\text{argmax}_{(\vec{w}^*, b^*)} \leq \frac{2}{\|\vec{w}\|}$$

such that

maximise it

with condn →

$$y_i (\vec{w}_i \cdot \vec{x}_i + b) \geq 1$$

One issue is Hard-margin svm

* need perfect linear data

solution of hard margin is svm is
soft margin svm

Soft margin svm

↳ don't ignore outlier, it also include it

maximise $f(x) (=)$ min $\frac{1}{f(x)}$

$$\arg \min_{(w^*, b^*)} \frac{\|w\|^2}{2} + \sum_{i=1}^n \zeta_i \times C$$

$\downarrow$
zeta

svm error

a) margin error

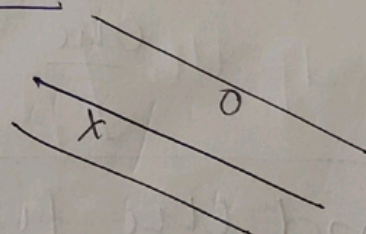
$$\frac{\|w\|^2}{2}$$

b) classification error

$$C \sum_{i=1}^n \zeta_i$$

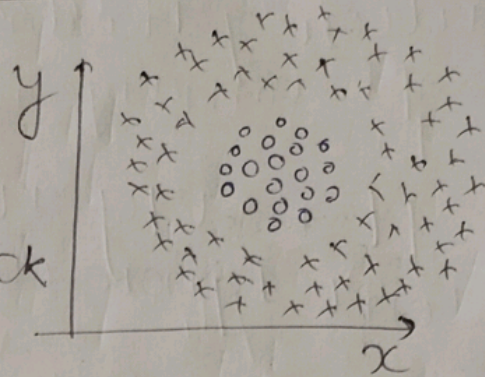
↳ hyper parameter.

$\zeta_i \approx 0$ for correctly classified points



Kernel & Trick SVM

SVM work with
non-linear data so
How

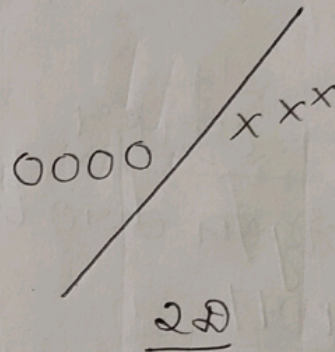


Ans:) using Kernel trick
Intuition

Convert the low dimension
data into the high dimension
data so, that it can be linear
separable.

eg:) 1D data into 2D data

eg:) 000 x x x x 000
 one D
 one



find the function
which convert transform
the low dimension to high
dimension and the function
called Kernels

types of Kernel

a) RBF

b) polynomial

c) sigmoid

d) linear

c) sigmoid.

Hyper parameter name is "Gridsearchcv"
